Ferdinando **Fioretto**

Copenhaver Fellow and Associate Professor

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Rice Hall 307, Computer Science, University of Virginia, Charlottesville - VA 22903 - U.S.A.

Research Interests : Artificial Intelligence | Responsible AI | AI for Science and Engineering

Professional Experience

present Jun. 2026	**University of Virginia**, *Computer Science*, Charlottesville, VA Copenhaver Fellow and Associate Professor
May 2026 Aug. 2023	**University of Virginia**, *Computer Science*, Charlottesville, VA Assistant Professor
Jul. 2023 Jan. 2020	**Syracuse University**, *Electrical Engineering & Computer Science*, Syracuse, NY Assistant Professor

Education and Training

Dec. 2019 Sep. 2018	**Georgia Institute of Technology**, *School of Industrial and System Engineering*, Atlanta, GA Post-doctoral Researcher
Dec. 2018 Sep. 2016	**University of Michigan**, *Industrial and Operations Engineering*, Ann Arbor, MI Research Fellow
Aug. 2016 Jan. 2012	**University of Udine** [1], *Computer Science*, Udine, IT Ph.D. in Computer Science
Dec. 2011 Aug. 2010	**New Mexico State University**, *Computer Science*, Las Cruces, NM MS. in Computer Science
Nov. 2009	**University of Parma**, *Computer Science & Mathematics*, Parma, IT BS. in Computer Science

Selected Honors and Awards

2026 **Copenhaver Fellow**, University of Virginia, endowed position. Link
- Faculty achievement and exceptional research trajectory.

2026 **Best Paper Award**, Institute of Industrial and System Engineers (IISE DAIS). Link
- The purpose of this competition is to honor outstanding papers in research and practice of data analytics and information systems published in any venue in the 2025-2026 year.

2025 **Academic Grant Program Award**, NVIDIA. Link
- The NVIDIA Academic Grant Program is a competitive program providing world-class computing access and resources to selected researchers.

2025 **DARPA Disruptive Idea Award**, NeuS 2025. Link
- This award supports bold, high-risk ideas that challenge conventional thinking in AI, even if the methodologies or experimental results are still in early stages.

2025 **Fellowship in AI Research**, LaCross Institute. Link
- Project name : **"Privacy and Fairness in AI Pipelines : From Data Collection to Decision-Making"**.
  The LaCross AI Institute was established in 2024 with the mission to make the world a better place through the responsible use of AI in business by developing leaders who can manage AI businesses and solutions, guided by ethics, values, and the advancement of human well-being. The Fellowships in AI Research program, supports scholars engaging in research that has beneficial practical societal outcomes and provides the foundationon for substantive future work.

2022 **Google Research Scholar Award**, Google (Privacy). Link

1. Dual degree with New Mexico State University

› Project name : **"Equity of Differentially Private Decision Processes"**.
The Research Scholar Program provides unrestricted gifts to support research at institutions around the world, and is focused on funding world-class research conducted by early-career professors.

2022 **Caspar Bowden PET Award**, Privacy Enhancing Technologies (PETs). Link

› The Caspar Bowden PET award for Outstanding Research in Privacy Enhancing Technologies is presented annually to researchers whose work makes an outstanding contribution to the theory, design, implementation, or deployment of privacy enhancing technology. The 2022 award was selected among all qualifying papers (published in **any** venue in the years 2020–2021).
The award letter reads : "Your paper Decision Making with Differential Privacy under the Fairness Lens received the award especially for advancing the understanding of DP and fairness trade-offs in decision making, providing a theoretical framework and exploring a highly relevant practical problem."

2022 **NSF CAREER Award**, National Science Foundation. Press

› Project name : **"End-to-end Constrained Optimization Learning"**. The Faculty Early Career Development (CAREER) Program is a Foundation-wide activity that offers the National Science Foundation's most prestigious awards in support of early-career faculty who have the potential to serve as academic role models in research and education and to lead advances in the mismission of their department or organization. Activities pursued by early-career faculty should build a firm foundation for a lifetime of leadership in integrating education and research.

2022 **Amazon Research Award**, Amazon – AWS AI (Responsible AI). Press

› Project name : **"Toward Understanding the Unintended Disparate Impacts of Private ML Systems"**.
The Amazon Research Awards is a competitive global program which offers unrestricted funds and AWS Promotional Credits to support research at academic institutions and non-profit organizations in areas that align Amazon's mission to advance science.

2022 **Best Paper Award**, IEEE Transaction of Power Systems. Link

› For paper : "Differentially Private Optimal Power Flow for Distribution Grids".
This highly selective award was assigned to eight out of all IEEE-TPWRS papers published in 2019–2021.

2022 **Early Career Spotlight**, International Joint Conference on Artificial Intelligence (IJCAI). Link

› Accompanying paper : "Integrating Machine Learning and Optimization to Boost Decision Making".
The IJCAI Early Career Spotlight talks are aimed at providing an accessible introduction to the research directions of some of the most active early career researchers in AI. The talks are by invitation, based on nominations from the IJCAI program committee.

2021 **Early Career Researcher Award**, Association for Constraint Programming. Link

› The Early Career Research Award is assigned by the Association for Constraint Programming to early career researchers for their contributions to constrained optimization.
In particular, this *inaugural* award was given "for contribution to constraint programming and, in particular, fundamental advances in distributed constraint satisfaction, constraint-based differential privacy, fairness in artificial intelligence, and their applications in energy, mobility, and census data."

2021 **Mario Gerla Young Investigator Award**, ISSNAF. Press

› Established by the Gerla family in 2019 in memory of Dr. Mario Gerla, professor of Computer Science at UCLA, the Italian Scientists and Scholars in North America Foundation confers the *Young Investigator Awards* every year to outstanding, early-career, Italian researchers working in North America, in recognition of their significant and innovative contributions to computer science. The award is conferred in coordination with the Italian Embassy in US.

2021 **Best Paper Award**, IEEE Transaction of Power Systems. Link

› For paper : "Privacy-Preserving Power System Obfuscation: A Bilevel Optimization Approach".
This highly selective award was assigned to seven out of all IEEE-TPWRS papers published in 2018–2020.

2017 **Best AI Dissertation Award**, AI*IA. Press

› For Thesis "**Exploiting the Structure of Distributed Constraint Optimization Problems with Applications in Smart Grids.**"
The "Marco Cadoli' 'Best AI dissertation is assigned by the Italian Association for Artificial Intelligence (AI*IA) to a Ph.D. doctor who have obtained the title in an Italian University based on the quality and impact of the thesis work.

Other Awards

2026 **Top Area Chair (Top AC) Award**, Conference on Neural Information Processing Systems (NeurIPS).
2025 **Best Paper Award**, NeurIPS AI4D3 workshop. Link
2025 **Best Paper Award**, AAAI colorai workshop. Link
2025 **Outstanding Research Faculty Award (department nomination)**, University of Virginia. Link
2023 **ICLR Notable Reviewer Award**, International conference on Learning Representations (ICLR). Link
2023 **NMSU CS Star Award**, New Mexico State University (NMSU). Link
2022 **Best Paper Award nomination**, Conference on Neural Information Processing Systems (NeurIPS). Link

2022 **Top Reviewer Award**, Conference on Neural Information Processing Systems (NeurIPS). Link
2021 **Outstanding Reviewer Award**, Conference on Neural Information Processing Systems (NeurIPS). Link
2020 **Differentially Private Temporal Map Challenge Award, $5000**, NIST. Press
2017 **Most Visionary Workshop Paper Award**, International Conference of Autonomous Agents and Multiagent Systems (AAMAS). Link
2016 **Top Graduate Student Honor's Cord**, NMSU.
2014 **Outstanding Research Assistant Award**, Computer Science, NMSU.
2014 **Outstanding Teaching Assistant Nomination**, NMSU.
2013 **Best Student Paper Award**, Computational Methods in System Biology (CMSB). Link
2013 **Ph.D. Scholarship Award (∼$50,000)**, University of Udine.
2013 **Outstanding Teaching Assistant Award**, Computer Science, NMSU.
2013 **Computer Science Scholarship ($1500)**, NMSU.
2012 **Honors Graduate Recognition for Outstanding Academic Success**, NMSU.
2008 **Erasmus Scholarship ($\sim \$14,000$)**, University of Leeds.

Publications

Summary : › *94* Conference papers† › *17* Journal articles › *41* Workshop papers† › *4* Editorial articles › *3* Book chapters › *1* Book › *20* Preprints and technical reports

Total citations : 4843 **H-index** : 32 Google Scholar

† : In computer science conferences are the main publication venues. Acceptance is highly competitive.

› Authors order usually follows contribution, where students are first authors and faculty advisors are listed last to reflect their supervisory role. In the following, when the symbol $[\alpha/\beta]$ is used, authors are ordered alphabetically. When the symbol $[c]$ is used, author ordered follows contribution, regardless of their advisor or seniority role.

› Names of students I supervise(d) are appended with symbol .

› Award winning papers are highlighted with symbol .

Books

1. Differential Privacy in Artificial Intelligence: From Theory to Practice. [] **Ferdinando Fioretto** and Pascal Van Hentenryck. Now Publisher, 2025.

Rigorously Peer Reviewed Conferences

In computer science, conference proceedings serve as the primary outlet for disseminating significant research contributions.

94. Chance-constrained Flow Matching for High-Fidelity Constraint-aware Generation. [] Jinhao Liang, Yixuan Sun, Anirban Samaddar, Sandeep Madireddy, **Ferdinando Fioretto**. **Conference on Neural Information Processing Systems (NeurIPS)**, 2026.
[Acceptance Rate: 7,900 / 30,709 = 25.7%].

93. Simulation-Informed Diffusion for Decentralized Multi-robot Motion Planning. [] Jinhao Liang, Sven Koenig, **Ferdinando Fioretto**. **Conference on Neural Information Processing Systems (NeurIPS)**, 2026.
[Acceptance Rate: 7,900 / 30,709 = 25.7%].

92. Drift Flow Matching. [] Chenrui Ma, Xi Xiao, Lin Zhao, Tianyang Wang, **Ferdinando Fioretto**, Yanning Shen. **Conference on Neural Information Processing Systems (NeurIPS)**, 2026.
[Acceptance Rate: 7,900 / 30,709 = 25.7%].

91. Constrained Code Generation with Discrete Diffusion. [] Lize Shao, Michael Cardei, Zichen Xie, **Ferdinando Fioretto**, Wenxi Wang. **Conference on Neural Information Processing Systems (NeurIPS)**, 2026.
[Acceptance Rate: 7,900 / 30,709 = 25.7%].

90. Colosseum: Auditing Collusion in Cooperative Multi-Agent Systems. [] Mason Nakamura, Abhinav Kumar, Saswat Das, Sahar Abdelnabi, Saaduddin Mahmud, **Ferdinando Fioretto**, Shlomo Zilberstein, Eugene Bagdasarian. **Conference on Neural Information Processing Systems (NeurIPS)**, Evaluations and Datasets Track, 2026.
[Acceptance Rate: 971 / 3,757 = 25.8%].

89. Search-Augmented Masked Diffusion Models for Constrained Generation. [] Huu Binh Ta, Michael Cardei, Alvaro Velasquez, **Ferdinando Fioretto**. **Conference on Neural Information Processing Systems (NeurIPS)**, 2026.
[Acceptance Rate: 7,900 / 30,709 = 25.7%].

88. FREESTYLE: An Anchor-Free Mechanism for Training-Free Style-Aligned Image Generation. [🔗] Minseok Oh, Jihun Park, Jongmin Gim, Minwoo Choi, Kyoungmin Lee, **Ferdinando Fioretto**, and Sunghoon Im. **The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)**, pages 3779–3788, 2026.
[Acceptance Rate: 4,090 / 16,092 = 25.42%].

87. Constrained Diffusion for Protein Design with Hard Structural Constraints. [🔗] Jacob K. Christopher, Austin Seamann, Jingyi Cui, Sagar Khare, **Ferdinando Fioretto**. **International Conference on Learning Representations (ICLR)**, 2026.
[Acceptance Rate: 5,320 / 19000 = 28.0%].

86. 🏆 Gen-DFL: Decision-Focused Generative Learning for Robust Decision Making. [🔗] Prince Zizhuang Wang, Jinhao Liang, Shuyi Chen, **Ferdinando Fioretto**, Shixiang Zhu. **International Conference on Learning Representations (ICLR)**, 2026.
[Acceptance Rate: 5,320 / 19000 = 28.0%]. Winner of the IISE DAIS best paper competition, 2026 .

85. 🏆 SpecDiff-2: Scaling Diffusion Drafter Alignment For Faster Speculative Decoding. [🔗] Jameson Sandler, Jacob K. Christopher, Thomas Hartvigsen, **Ferdinando Fioretto**. **In Conference on Machine Learning and Systems (MLSys)**, 2026.
[Acceptance Rate: 133/504 = 26.4%]. [Oral] .

84. 🏆 A Method for Learning to Solve Parametric Bilevel Optimization with Coupling Constraints. [🔗] James Kotary, Himanshu Sharma, Ethan King, Draguna L Vrabie, **Ferdinando Fioretto**, Jan Drgona. **Annual Learning for Dynamics & Control Conference (L4DC)**, 2026.
[Acceptance Rate: Unknown]. [Oral] .

83. 🏆 SoK: Data Minimization in Machine Learning. [🔗] Robin Staab, Nikola Jovanović, Kimberly Mai, Prakhar Ganesh, Martin Vechev, **Ferdinando Fioretto**, Matthew Jagielski. **IEEE Secure and Trustworthy Machine Learning Conference**, 2026.
[Acceptance Rate: 61 / 233 = 26.2%]. [Oral] .

82. Discrete-Guided Diffusion for Scalable and Safe Multi-Robot Motion Planning. [🔗] Jinhao Liang, Sven Koenig, **Ferdinando Fioretto**. **AAAI Conference on Artificial Intelligence (AAAI)**, 2026.
[Acceptance Rate: 4,167 / 23,680 = 17.6%].

81. 🏆 Training-Free Constrained Generation With Stable Diffusion Models. [🔗] Stefano Zampini, Jacob K Christopher, Luca Oneto, Davide Anguita, **Ferdinando Fioretto**. **Conference on Neural Information Processing Systems (NeurIPS)**, 2025.
[Acceptance Rate: 5290 / 21575 = 24.5%]. [Spotlight] (3.18% of the submitted papers).

80. Constrained Discrete Diffusion. [🔗] Michael Cardei, Jacob K Christopher, Thomas Hartvigsen, Bhavya Kailkhura, **Ferdinando Fioretto**. **Conference on Neural Information Processing Systems (NeurIPS)**, 2025.
[Acceptance Rate: 5290 / 21575 = 24.5%].

› Also appeared in the **AI4D3 Workshop at NeurIPS**, 2025. 🏆[Oral] (6% of the accepted papers (36% acceptance rate)).

79. Robust Bidding Strategies in Local Energy Markets. Jinhao Liang, **Ferdinando Fioretto**, Chenye Wu. **IEEE PES General Meeting**, pages 1–5, 2025.
[Acceptance Rate: Unknown].

78. Privacy-Preserving Convex Optimization: When Differential Privacy Meets Stochastic Programming. [🔗] Vladimir Dvorkin, **Ferdinando Fioretto**, Pascal Van Hentenryck, Pierre Pinson, Jalal Kazempour. **IEEE Conference on Decision and Control (CDC)**, 2025.
[Acceptance Rate: 1,328 / 2,318 = 57.3%].

77. Simultaneous Multi-Robot Motion Planning with Projected Diffusion Models. [🔗] Jinhao Liang, Jacob K. Christopher, Sven Koenig, **Ferdinando Fioretto**. **International Conference on Machine Learning (ICML)**, 2025.
[Acceptance Rate: 3,260 / 12,107 = 26.9%].

76. 🏆 Neuro-symbolic Generative Diffusion Models for Physically Grounded, Robust, and Safe Generation. [🔗] Jacob K. Christopher, Michael Cardei, Jinhao Liang, **Ferdinando Fioretto**. **International Conference on Neuro-symbolic Systems (NeuS)**, 2025.
[Acceptance Rate: 39 / 100 = 39%]. [Oral (∼10% of accepted papers) & DARPA Disruptive Idea Award] .

75. The Data Minimization Principle in Machine Learning. [🔗] Prakhar Ganesh, Cuong Tran, Reza Shokri, **Ferdinando Fioretto**. **ACM Conference on Fairness, Accountability, and Transparency (FAccT)**, 2025.

[Acceptance Rate: 812 / 3,031 = 26.8%].

74. Learning To Solve Differential Equation Constrained Optimization Problems. [🔗] Vincenzo Di Vito, Mostafa Mohammadian, Kyri Baker, **Ferdinando Fioretto**. **International Conference on Learning Representations (ICLR)**, 2025.
[Acceptance Rate: 3,704 / 11,672 = 31.73%]. [Spotlight] (3.26% of the submitted papers).

73. Speculative Diffusion Decoding: Accelerating Language Generation through Diffusion. [🔗] Jacob K. Christopher, Michael Cardei, Brian R Bartoldson, Bhavya Kailkhura, **Ferdinando Fioretto**. **North American Chapter of the Association for Computational Linguistics (NAACL)**, 2025.
[Acceptance Rate: 3,185 / 7,719 = 22.5%]. [oral] (14.9% of the submitted papers).

› Also appeared in [w28] at **NeurIPS**, 2024. [**Accepted Rate** : 7.7%].

72. Differentially Private Data Release on Graphs: Inefficiencies and Unfairness. [🔗] [α/β] **Ferdinando Fioretto**, Diptangshu Sen, Juba Ziani. **International Conference on Artificial Intelligence and Statistics (AISTATS)**, 2025.
[Acceptance Rate: 583 / 1,861 = 31.3%].

71. Fairness Issues and Mitigations in (Differentially Private) Socio-demographic Data Processes. [🔗] Joonhyuk Ko, Juba Ziani, Saswat Das, Matt Williams, **Ferdinando Fioretto**. **AAAI Conference on Artificial Intelligence (AAAI)**, 2025.
[Acceptance Rate: 89/469 = 23.4%]. [Oral] 1.9% of the submitted papers).

› Also appeared in [w33] at **AAAI**, 2024. [Oral] (7.6% of the accepted papers).

70. FairDP: Certified Fairness with Differential Privacy. [🔗] [c] Khang Tran, **Ferdinando Fioretto**, Issa Khalil, My T. Thai, NhatHai Phan. **IEEE Secure and Trustworthy Machine Learning Conference (SaTML)**, 2025.
[Acceptance Rate: 53/180 = 29.4%].

69. Constrained Synthesis with Projected Diffusion Models. [🔗] Jacob K. Christopher, Stephen Baek, **Ferdinando Fioretto**. **Conference on Neural Information Processing Systems (NeurIPS)**, 2024.
[Acceptance Rate: 25.8%].

› Also appeared in [w27,w29] at **NeurIPS**, 2024. [Oral] (6% of the accepted papers (39% acceptance rate)).

68. Metric Learning to Accelerate Convergence of Operator Splitting Methods for Differentiable Parametric Programming. [🔗] [c] Ethan King, James Kotary, **Ferdinando Fioretto**, Jan Drgona. **63rd IEEE Conference on Decision and Control (CDC)**, 2024.
[Acceptance Rate: 56.7%].

67. Predict-Then-Optimize by Proxy: Learning Joint Models of Prediction and Optimization. [🔗] James Kotary, Vincenzo Di Vito, Jacob K. Christopher, Pascal Van Hentenryck, **Ferdinando Fioretto**. **European Conference of Artificial Intelligence (ECAI)**, 2024.
[Acceptance Rate: 23.3%].

66. On The Fairness Impacts of Hardware Selection in Machine Learning. [🔗] Sree Harsha Nelaturu, Nishaanth Kanna Ravichandran, Cuong Tran, Sara Hooker, **Ferdinando Fioretto**. **International Conference on Machine Learning (ICML)**, 2024.
[Acceptance Rate: 27.5%].

65. Disparate Impact on Group Accuracy of Linearization for Private Inference. [🔗] Saswat Das, Marco Romanelli, **Ferdinando Fioretto**. **International Conference on Machine Learning (ICML)**, 2024.
[Acceptance Rate: 27.5%].

64. End-to-End Learning for Fair Multiobjective Optimization Under Uncertainty. [🔗] My H. Dinh, James Kotary, **Ferdinando Fioretto**. **Conference on Uncertainty in Artificial Intelligence (UAI)**, 2024.
[Acceptance Rate: 27.0%].

63. Fairness Increases Adversarial Vulnerability. [🔗] Cuong Tran, Keyu Zhu, Pascal Van Hentenryck, **Ferdinando Fioretto**. **International Joint Conference on Artificial Intelligence (IJCAI)**, 2024.
[Acceptance Rate: 13.9%].

62. Learning Fair Ranking Policies via Differentiable Optimization of Ordered Weighted Averages. [🔗] My H. Dinh, James Kotary, **Ferdinando Fioretto**. **ACM Conference on Fairness, Accountability, and Transparency (FAccT)**, 2024.
[Acceptance Rate: 24.3%].

61. Finding ϵ and δ of Traditional Disclosure Control Systems. [🔗] [c] **Ferdinando Fioretto**, Keyu Zhu, Pascal Van Hentenryck, Saswat Das, Christine Task. **AAAI Conference on Artificial Intelligence (AAAI)**, 2024.
[Acceptance Rate: 23.75%].

60. Data Minimization at Inference Time. [🔗] Cuong Tran and **Ferdinando Fioretto**. **Conference on Neural Information Processing Systems (NeurIPS)**, 2023.
[Acceptance Rate: 23.0%].

59. Price-Aware Deep Learning for Electricity Markets. [🔗] Vladimir Dvorkin and **Ferdinando Fioretto**. **Tackling Climate Change with Machine Learning, at NeurIPS**, 2023.
[Acceptance Rate: 35.0%].

58. Folded Optimization for End-to-End Model-Based Learning. [🔗] James Kotary, My H. Dinh, **Ferdinando Fioretto**. **International Joint Conference on Artificial Intelligence (IJCAI)**, 2023.
[Acceptance Rate: 15.0%].

57. SF-PATE: Scalable, Fair, and Private Aggregation of Teacher Ensembles. [🔗] James Kotary, Vincenzo Di Vito, **Ferdinando Fioretto**, Pascal Van Hentenryck. **International Joint Conference on Artificial Intelligence (IJCAI)**, 2023.
[Acceptance Rate: 15.0%].

56. End-to-End Combinatorial Ensemble Learning. [🔗] James Kotary, Vincenzo Di Vito, **Ferdinando Fioretto**. **International Joint Conference on Artificial Intelligence (IJCAI)**, 2023.
[Acceptance Rate: 15.0%].

55. On the Fairness Impacts of Private Ensembles Models. [🔗] Cuong Tran, **Ferdinando Fioretto**. **International Joint Conference on Artificial Intelligence (IJCAI)**, 2023.
[Acceptance Rate: 15.0%].

54. Load Encoding for Learning AC-OPF. [🔗] [c] Terrence W.K. Mak, **Ferdinando Fioretto**, Pascal Van Hentenryck. Proceedings of the **IEEE PES General Meeting (PES)**, 2023.
[Acceptance Rate: Unknown].

53. An Analysis of the Reliability of AC Optimal Power Flow Deep Learning Proxies. [🔗] [c] My H. Dinh, **Ferdinando Fioretto**, Mostafa Mohammadian, Kyri Baker. **IEEE PES Innovative Smart Grid Technologies**, 2023.
[Acceptance Rate: unknown].

52. End-to-End Optimization and Learning for Multiagent Ensembles. [🔗] James Kotary, Vincenzo Di Vito, **Ferdinando Fioretto**. **International Conference on Autonomous Agents and Multiagent Systems (AAMAS)**, 2023.
[Acceptance Rate: 40.0%].

51. 🏆 Pruning has a disparate impact on model accuracy. [🔗] Cuong Tran, **Ferdinando Fioretto**, Jung-Eun Kim, Rakshit Naidu. **Conference on Neural Information Processing Systems (NeurIPS)**, 2022.
[Acceptance Rate: 25.6%]. [Spotlight] (∼3% of all paper submissions.).

50. Post-processing of Differentially Private Data: A Fairness Perspective. [🔗] [c] Keyu Zhu, **Ferdinando Fioretto**, Pascal Van Hentenryck. **International Joint Conference on Artificial Intelligence (IJCAI)**, 2022.
[Acceptance Rate: 15.0%].

49. Differential Privacy and Fairness in Decisions and Learning Tasks: A Survey. [🔗] [c] **Ferdinando Fioretto**, Cuong Tran, Keyu Zhu, Pascal Van Hentenryck. **International Joint Conference on Artificial Intelligence (IJCAI)**, 2022.
[Acceptance Rate: 18.0% (survey track)].

48. 🏆 Integrating Machine Learning and Optimization to Boost Decision Making. [🔗] **Ferdinando Fioretto**. **International Joint Conference on Artificial Intelligence (IJCAI)**, 2022.
[Acceptance Rate: Invited]. [Early Career Spotlight] (Accompanying paper.).

47. End-to-end Learning for Fair Ranking Systems. [🔗] [c] James Kotary, **Ferdinando Fioretto**, Pascal Van Hentenryck, Ziwei Zhu. **ACM Web Conferences (WWW)**, 2022.
[Acceptance Rate: 17.0%].

46. Fast Approximations for Job Shop Scheduling: A Lagrangian Dual Deep Learning Method. [🔗] [c] James Kotary, **Ferdinando Fioretto**, Pascal Van Hentenryck. **AAAI Conference on Artificial Intelligence (AAAI)**, 2022.
[Acceptance Rate: 15.0%].

45. Differentially-Private Heat and Electricity Markets Coordination. [🔗] Lesia Mitridati, Emma Romei, Gabriela Hug, **Ferdinando Fioretto**. **International Conference on Probabilistic Methods Applied to Power Systems (PMAPS)**, 2022.

[Acceptance Rate: Unknown].

44. Learning Solutions for Intertemporal Power Systems Optimization with Recurrent Neural Networks. [🔗] [c] Mostafa Mohammadian, Kyri Baker, My H. Dinh, **Ferdinando Fioretto**. **International Conference on Probabilistic Methods Applied to Power Systems (PMAPS)**, 2022.
[Acceptance Rate: Unknown].

43. Differentially Private Deep Learning under the Fairness Lens. [🔗] Cuong Tran, My H. Dinh, **Ferdinando Fioretto**. **Conference on Neural Information Processing Systems (NeurIPS)**, 2021.
[Acceptance Rate: 26.0%].

42. Learning Hard Optimization Problems: A Data Generation Perspective. [🔗] James Kotary, [c] **Ferdinando Fioretto**, Pascal Van Hentenryck. **Conference on Neural Information Processing Systems (NeurIPS)**, 2021.
[Acceptance Rate: 26.0%].

41. 🏆 Decision Making with Differential Privacy under the Fairness Lens. [🔗] [c] Cuong Tran, **Ferdinando Fioretto**, Pascal Van Hentenryck, Zhiyan Yao. **International Joint Conference on Artificial Intelligence (IJCAI)**, 560–566, 2021.
[Acceptance Rate: 13.9%]. [Winner of the 2022 Caspar Bowden PET Award] (Selected among all papers about Privacy Enhancing Technologies published in international conferences between 2020–2022.).

40. End-to-End Constrained Optimization Learning: A Survey. [🔗] [c] James Kotary, **Ferdinando Fioretto**, Pascal Van Hentenryck, Bryan Wilder. **International Joint Conference on Artificial Intelligence (IJCAI)**, 4475–4482, 2021.
[Acceptance Rate: 30.1%].

39. Bias and Variance of Post-processing in Differential Privacy. [🔗] Keyu Zhu, Pascal Van Hentenryck, **Ferdinando Fioretto**. **AAAI Conference on Artificial Intelligence (AAAI)**, 11177–11184, 2021.
[Acceptance Rate: 21.0%].

38. Differentially Private and Fair Deep Learning: A Lagrangian Dual Approach. [🔗] Cuong Tran, [c] **Ferdinando Fioretto**, Pascal Van Hentenryck. **AAAI Conference on Artificial Intelligence (AAAI)**, 9932–9939, 2021.
[Acceptance Rate: 21.0%].

37. A Privacy-Preserving and Accountable Multi-agent Learning Framework. [🔗] Anudit Nagar, Cuong Tran, **Ferdinando Fioretto**. **International Conference on Autonomous Agents and Multiagent Systems (AAMAS)**, 1605–1606, 2021.
[Acceptance Rate: 40.0%].

36. Constrained-based Differential Privacy. [🔗] **Ferdinando Fioretto. International Conference on Principles and Practice of Constraint Programming (CP), 1868–8969, 2021.**
[Acceptance Rate: Invited].

35. Differentially Private Optimal Power Flow for Distribution Grids. [🔗] [c] Vladimir Dvorkin, **Ferdinando Fioretto**, Pascal Van Hentenryck, Jalal Kazempour, Pierre Pinson. **IEEE PowerTech**, 2021.
[Acceptance Rate: Unknown].

34. A Lagrangian Dual Framework for Deep Neural Networks with Constraints. [🔗] [c] **Ferdinando Fioretto**, Pascal Van Hentenryck, Terrence W.K. Mak, Cuong Tran, Federico Baldo, Michele Lombardi. **European Conference on Machine Learning (ECML)**, 18–135, 2020.
[Acceptance Rate: 19.0%].

33. Differential Privacy Stackebelg Games. [🔗] [c] **Ferdinando Fioretto**, Lesia Mitridati, Pascal Van Hentenryck. **International Joint Conference on Artificial Intelligence (IJCAI)**, 3480–3486, 2020.
[Acceptance Rate: 12.6%].

32. 🏆 OptStream: Releasing Time Series Privately. [🔗] **Ferdinando Fioretto**, Pascal Van Hentenryck. **International Joint Conference on Artificial Intelligence (IJCAI)**, 5135–5139, 2020.
[Acceptance Rate: [Invited to the IJCAI journal track]]. (selected papers only.) .

31. Privacy-Preserving Obfuscation for Distributed Power Systems. [🔗] [c] Terrence W.K. Mak, **Ferdinando Fioretto**, Pascal Van Hentenryck. **Power Systems Computation Conference (PSCC)**, 2020.
[Acceptance Rate: 20.5%].

30. Predicting AC Optimal Power Flows: Combining Deep Learning and Lagrangian Dual Methods. [🔗] **Ferdinando Fioretto**, Terrence W.K. Mak, Pascal Van Hentenryck. **AAAI Conference on Artificial Intelligence (AAAI)**, pages 630–637, 2020.
[Acceptance Rate: 20.6%].

29. The Smart Appliance Scheduling Problem: A Bayesian Optimization Approach. [🔗] Atena Tabakhi, William Yeoh, **Ferdinando Fioretto**. **International Conference on Principles and Practice of Multi-Agent Systems (PRIMA)**, 100–115, 2020.
[Acceptance Rate: 38.0%].

28. Privacy-Preserving Federated Data Sharing. [🔗] **Ferdinando Fioretto**, Pascal Van Hentenryck. **International Conference on Autonomous Agents and Multiagent Systems (AAMAS)**, pages 638–646, 2019.
[Acceptance Rate: 24.0%].

27. Privacy-Preserving Obfuscation of Critical Infrastructure Networks. [🔗] **Ferdinando Fioretto**, Terrence W.K. Mak, Pascal Van Hentenryck. **International Joint Conference on Artificial Intelligence (IJCAI)**, pages 1086–1092, 2019.
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10. NeuroFilter: Privacy Guardrails for Conversational LLM Agents. [🔗] Saswat Das, **Ferdinando Fioretto**. **CoRR abs**/2601.14660, 2026.
9. The Disparate Impacts of Speculative Decoding. [🔗] Jameson Sandler, Ahmet Üstün, Marco Romanelli, Sara Hooker, **Ferdinando Fioretto**. **CoRR abs**/2510.02128, 2025.
8. Learning to Solve Optimization Problems Constrained with Partial Differential Equations. [🔗] Yusuf Guven, Vincenzo Di Vito, **Ferdinando Fioretto**. **CoRR abs**/2509.24573, 2025.
7. Beyond Jailbreaking: Auditing Contextual Privacy in LLM Agents. [🔗] Saswat Das, Jameson Sandler, **Ferdinando Fioretto**. **CoRR abs**/2506.10171, 2025.
6. Optimal Allocation of Privacy Budget on Hierarchical Data Release. [🔗] Joonhyuk Ko, Juba Ziani, **Ferdinando Fioretto**. **CoRR abs**/2505.10871, 2025.

5. How Can We Ground Frontiers Models for Biological Research to Domain Knowledge, First Principles, and Constraints?. [🔗] **Ferdinando Fioretto**, Bhavya Kailkhura. **Lawrence Livermore National Laboratory white paper**, LLNL-CONF-871579, 2025.
4. End-to-End Optimization and Learning of Fair Court Schedules. [🔗] My H. Dinh, James Kotary, Lauryn P. Gouldin, William Yeoh, **Ferdinando Fioretto**. **CoRR abs**/2410.17415, 2024.
3. Low-rank finetuning for LLMs: A fairness perspective. [🔗] Saswat Das, Marco Romanelli, Cuong Tran, Zarreen Reza, Bhavya Kailkhura, **Ferdinando Fioretto**. **CoRR abs**/2405.18572, 2024.
2. Learning Constrained Optimization with Deep Augmented Lagrangian Methods. [🔗] James Kotary, **Ferdinando Fioretto**. **CoRR abs**/2403.03454, 2024.
1. Analyzing and Enhancing the Backward-Pass Convergence of Unrolled Optimization. [🔗] James Kotary, Jacob K. Christopher, My H Dinh, and **Ferdinando Fioretto**. **CoRR abs**/2312.17394, 2023.

Archived and Extended Versions of Published Papers

12. Context-Aware Differential Privacy for Language Modeling. [🔗] My H. Dinh, **Ferdinando Fioretto**. **CoRR abs**/2301.12288, 2023.
11. Deadwooding: Robust Global Pruning for Deep Neural Networks. [🔗] Sawinder Kaur, **Ferdinando Fioretto**, Asif Salekin. **CoRR abs**/2202.05226, 2022.
10. Towards Understanding the Unreasonable Effectiveness of Learning AC-OPF Solutions. [🔗] My H. Dinh, **Ferdinando Fioretto**, Mostafa Mohammadian, Kyri Baker. **CoRR abs**/2111.11168, 2021.
9. Differentially Private Deep Learning under the Fairness Lens. [🔗] Cuong Tran, My H. Dinh, **Ferdinando Fioretto**. **CoRR abs**/2106.02674, 2021 (**extended NeurIPS-21 version**).
8. A Privacy-Preserving and Trustable Multi-agent Learning Framework. [🔗] Anudit Nagar, Cuong Tran, **Ferdinando Fioretto**. **CoRR abs**/2106.01242, 2021. (**extended AAMAS-21 version**).
7. End-to-End Constrained Optimization Learning: A Survey. [🔗] James Kotary, **Ferdinando Fioretto**, Pascal Van Hentenryck, Bryan Wilder. **CoRR abs**/2103.16378, 2021. (**extended IJCAI-21 version**).
6. Load Embeddings for Scalable AC-OPF Learning. [🔗] Terrence W.K. Mak, **Ferdinando Fioretto**, Pascal Van Hentenryck. **CoRR abs**/2101.03973, 2021.
5. Bias and Variance of Post-processing in Differential Privacy. [🔗] Keyu Zhu, Pascal Van Hentenryck, **Ferdinando Fioretto**. **CoRR abs**/2010.04327, 2020 (**extended AAAI-21 version**).
4. High-Fidelity Machine Learning Approximations of Large-Scale Optimal Power Flow. [🔗] Minas Chatzos, **Ferdinando Fioretto**, Terrence W.K. Mak, Pascal Van Hentenryck. **CoRR abs**/2006.16356, 2020.
3. Differentially Private Convex Optimization with Feasibility Guarantees. [🔗] Vladimir Dvorkin, **Ferdinando Fioretto**, Pascal Van Hentenryck, Jalal Kazempour, Pierre Pinson. **CoRR abs**/2006.12338, 2020.
2. Predicting AC Optimal Power Flows: Combining Deep Learning and Lagrangian Dual Methods. [🔗] **Ferdinando Fioretto**, Terrence W.K. Mak, Pascal Van Hentenryck. **CoRR abs**/1909.10461, 2019 (**extended AAAI-20 version**).
1. Privacy-Preserving Obfuscation of Critical Infrastructure Networks. [🔗] **Ferdinando Fioretto**, Terrence W. K. Mak, Pascal Van Hentenryck. **CoRR abs**/1905.09778, 2019 (**extended IJCAI-19 version**).

Patents and Patent Applications

2. Constrained Sequence Generation Using Discrete Diffusion Models. **Ferdinando Fioretto**, Jacob K. Christopher, and Michael Cardei. U.S. Patent Application Serial No. 19/722,273, filed June 26, 2026.
1. Optimization-Based Differential Privacy. **Ferdinando Fioretto**, Chansoo Lee, and Pascal Van Hentenryck. University of Michigan technology disclosure, File No. 7707.

Teaching

Constraint-aware Generative AI for Science and Engineering (CS 6501), *University of Virginia*

Spring 2026	Course Evaluation : 4.7 (class), 4.7 (instructor) / 5.00

Responsible AI (CS 6501), *University of Virginia*

Spring 2025	Course Evaluation : 4.67 (class), 4.24 (instructor) / 5.00
Spring 2024	Course Evaluation : 4.8 (class), 4.75 (instructor) / 5.00

Artificial Intelligence (CS 4710), *University of Virginia*

Fall 2025	Course Evaluation : 4.70 (class), 4.60 (instructor) / 5.00
Fall 2024	Course Evaluation : 4.21 (class), 3.98 (instructor) / 5.00
Fall 2023	Course Evaluation : 4.33 (class), 4.50 (instructor) / 5.00

Security and Privacy of Machine Learning (CS 700), *Syracuse University*

Spring 2022	Course Evaluation : 4.93/5.00 (median 5.00)
Spring 2021	Course Evaluation : 4.46/5.00 (median 5.00)
Spring 2020	Course Evaluation : 4.55/5.00 (median 5.00)

Introduction to Artificial Intelligence (CIS 467), *Syracuse University*

Fall 2023	Course Evaluation : 4.15/5.00 (median 5.00)
Fall 2022	Course Evaluation : 4.45/5.00 (median 5.00)
Fall 2021	Course Evaluation : 4.48/5.00 (median 5.00)
Fall 2020	Course Evaluation : 4.56/5.00 (median 5.00)

Discrete Mathematics (CS 375), *Syracuse University*

Spring 2023	Course Evaluation : 4.60/5.00 (median 5.00)

Mentoring

Current PhD Students

University of Virginia, Computer Science. Dates indicate the start of the PhD program.

Student	PhD Start	Research Focus
Vincenzo Di Vito	Fall 2022	Physics-informed machine learning
Saswat Das	Fall 2023	Responsible AI; differential privacy; agentic AI
Jacob K. Christopher	Fall 2023	Generative AI for science; safety
Jinhao Liang	Fall 2024	Generative AI; differentiable optimization
Michael Cardei	Fall 2024	Constrained generative AI; diffusion language models
Huu Binh Ta	Fall 2025	Generative AI; AI for science
Zehua Wang	Fall 2025	Generative AI; privacy
Jameson Sandler	Fall 2025	Fairness; generative AI; optimization
Yusuf Güven	Fall 2026	Constraint-aware generative AI for engineering
Chenrui Ma	Fall 2026	Efficient generative modeling; trustworthy ML; multimodal AI

Current MS and BS Students

University of Virginia, Computer Science. Dates indicate the start of lab involvement.

Student	Program	Joined Lab	Research Interests / Area
Michael Martinez	MS	Spring 2026	Generative AI; constrained generation
Parvati Viswanathan	MS	Spring 2026	Agentic AI; AI safety
Ali Elsayyad	BS	Spring 2026	Generative AI; constrained generation
Jai Kanchan	BS	Spring 2026	Generative AI; constrained generation
Ani Venkatapuram	BS	Spring 2026	Generative AI; constrained generation
Michael Yoo Fatemi	BS	Fall 2025	Robotics; generative planning
Aarav Lodha	BS	Fall 2025	AI safety; technical alignment
Lauren LaPorta	BS	Spring 2025	Generative AI; AI governance
Connor Lewis	BS	Spring 2025	Generative AI; constrained generation
Alexander Yao	BS	Summer 2025	Generative AI; constrained generation

Former Students, Postdocs, and Visitors

Name	Lab Role / Degree	Period	Next / Current Position
Joonhyuk Ko	BS, UVA	2023–2026	PhD student, Princeton University (Fall 2026)
Peggy Cui	MS, UVA	Fall 2025	ML engineer, fAIshion

Name	Lab Role / Degree	Period	Next / Current Position
James Kotary	PhD, UVA	2020–2024	Postdoctoral researcher, Pacific Northwest National Laboratory
My Dinh	PhD, UVA	2021–2025	Data Scientist, Walmart
Cuong Tran	PhD, Syracuse; postdoc, UVA	2020–2023; 2023–2024	NLP Scientist, Dyania Health
Yehya Farhat	MS, Syracuse	2021–2022	PhD student, Rice University
Mehdi Taghizadeh	Postdoc; UVA Climate Fellow	Joined Fall 2025	AI research
Lea Demelius	Visiting PhD, TU Graz	Spring 2025	Affiliated researcher, TU Graz; PhD completed in 2026
Minseok Oh	Visiting undergraduate	Summer 2025	MS student, KAIST Robotics and Vision Lab
Stefano Zampini	Visiting PhD, U. Genoa	Summer 2024	PhD studies, University of Genoa
Razane Tajeddine	Visiting postdoc, U. Helsinki	Sep. 2023–Mar. 2024	Assistant Professor, Electrical and Computer Engineering, American University of Beirut
St John Grimbly	Visiting MS researcher	Spring 2023	PhD candidate, University of Cape Town
Jayanta Mandi	Visiting PhD, KU Leuven	Jun.–Sep. 2022	Postdoctoral Research Associate, Transport Data and AI, Loughborough University
Rakshit Naidu	MS research intern, CMU	Summer 2022	PhD student, Machine Learning, Georgia Tech
Kyle Beiter	Undergraduate NSF REU	Summer 2021	Data Scientist, Optiver

Mehdi Taghizadeh was co-advised with Negin Alemazkoor; research : learning to optimize and energy.

PhD Dissertations

- James Kotary : *Integrating Constrained Optimization with Machine Learning to Enhance Data-Driven Decision Making*.
- My Dinh : *Bridging Machine Learning and Optimization : Learning Fair and Scalable Problem Solving*.
- Cuong Tran : *The Interplay between Privacy and Fairness in Learning and Decision-making Problems*.

MS thesis. Yehya Farhat : *Surrogate ML Models for Optimization*.

Additional Undergraduate and High-School Research Mentees

UVA : University of Virginia; SU : Syracuse University; HS : high school.

Year	Mentees and Home Institutions
2026	**Junsung Lee**; **Yunho Cho** (summer research interns).
2025	**Xiaotian Gu**, **Natalia Wunder**, **Joseph Lee** (BS, UVA, Spring); **Kyle Briggs** (BS, UVA, Summer); **Aiden Want** (HS, Academy of Science, Summer).
2024	**Aran Jothi**, **Pranav Vaddepalli** (BS, UVA, Fall); **Shujun Xia** (BS, City University of Hong Kong, Summer); **Zarreen Reza** (BS, OpenMined, Spring).
2023–2024	**Catherine Smolka** (HS, Deep Run High School, Virginia).
2023	**Eric Nguyen** (BS, UVA, Fall); **Pranav Putta** (BS, Georgia Tech, Summer; NSF REU); **Winston Tsui**, **Zhongquan Cheng** (BS, SU, Summer).
2022	**Adya Parida** (BS, SU, Fall; NSF REU); **Deniz Gursoy** (HS, Fayetteville High School, Summer); **Saswat Das** (undergraduate researcher, NISER, Summer; now UVA PhD); **Utsav Pathak** (BS, Alliance University, Bengaluru, Summer); **Daiwei Shen** (BS, Northwestern, Summer); **Sunisth Kumar** (BS, Bennett University, Summer).
2021	**Shantanu Jhaveri** (BS, USC, Summer; NSF REU); **Dayong Gu**, **Guoliang Chen**, **Pradyumn Yadav** (BS, SU, Summer).
2020	**Anudit Nagar**, **Zhiyan Yao**, **Zifei Lu**, **Thomas Montfort**, **Cong Liu**, **Pratik Paranjape** (BS, SU, Summer); **Pavan Kumar Vaddineni** (BS, SU, Spring).
2016–2018	**William Kluegel** (BS, New Mexico State University, 2016–2018); **Lyndon Shi**, **Jiayu Chen** (BS, University of Michigan, 2018); **Eric Frechette** (BS, New Mexico State University, 2016).

PhD Dissertation Committee

Year	Students and Institutions
2026	**Ryan Chen** (Carnegie Mellon University); **Yucheng Fu** (University of Virginia); **Lea Demelius** (TU Graz).
2025	**Prince Wang** (Carnegie Mellon University); **Zihan QE**, **Rahul Ramesh Reddy**, **Xingbo Fu**, **Kai Chen**, **Anders Gyllenhoff**, **Hannah Cyberey**, **Chen Gong**, **Galen Harrison**, **Felipe Toledo** (University of Virginia); **Mostafa Mohammadian** (University of Colorado, Boulder).
2024	**Luca Giuliani**, **Eleonora Misino** (University of Bologna); **Guangtao Zheng** (University of Virginia).
2023	**Dung Nguyen**, **Elena Long** (University of Virginia); **Khang Tran** (New Jersey Institute of Technology); **Keyu Zhu** (Georgia Institute of Technology); **Adrià Fenoy Barcel** (University of Verona).
2022	**Jeroen Fransman** (Delft University of Technology); **Pegah Hozhabrierdi** (Syracuse University); **Carlos Pinzon** (École Polytechnique).
2021	**Baocheng Geng**, **Pranay Sharma** (Syracuse University).

Selected Student Awards and Research Experience

- Jacob K. Christopher, Michael Cardei, and Jinhao Liang : DARPA Disruptive Idea Award, NeuS 2025.
- Jacob K. Christopher and Michael Cardei : Best Student Paper Award, NeurIPS AI4D3 Workshop (2025); best paper, UVA LLM Workshop (2024); NAACL 2025 oral.
- Michael Cardei : NSF Graduate Research Fellowship (2025). Research internships : Visa (Summer 2025); Lawrence Livermore National Laboratory (2026).
- Jacob K. Christopher : NeurIPS 2025 spotlight; MLSys 2026 oral. Research experience : Lawrence Livermore National Laboratory (Summer 2024); NASA Langley (Oct. 2025–Jan. 2026); Cerebras Systems (Summer 2026).
- Saswat Das : Best Paper Award, AAAI CoLoRAI Workshop (2025); AAAI 2025 oral; ICML Gold Reviewer and NeurIPS Top Reviewer (2026). Internships : Pacific Northwest National Laboratory (Summer/Fall 2025 and 2026).
- Vincenzo Di Vito : ICLR 2025 spotlight. Internships : Los Alamos National Laboratory (Summers 2024, 2025); Google DeepMind (Fall 2025).
- Jinhao Liang : AAAI AI+ORMS Bridge and MAPF Workshop orals (2025). Internship : Argonne National Laboratory (Summer 2025).
- Chenrui Ma : UVA Computer Science Scholar Fellowship and NeurIPS Top Reviewer (2026). Student collaborator, Oak Ridge National Laboratory (2025–present).
- Joonhyuk Ko : NSF Graduate Research Fellowship (2026); CRA Outstanding Undergraduate Researcher Award finalist (2026), honorable mention (2025); UVA Louis T. Rader Undergraduate Research Award (2025); AAAI 2025 oral; Carnegie Mellon research internship (Summer 2025).
- Cuong Tran : Caspar Bowden PET Award (2022); NeurIPS 2022 Best Paper Award nomination.
- Jameson Sandler : highest distinction in CS, UVA.

Research Grants and Gifts

Summary : Total External : $4,586,403 ($3,234,636 as PI) Total Internal : $281,000

- Cerebras Systems - Contract, ($95,000) 9/26-8/27
  Project title : *Theory-grounded Methods for Efficient Discrete and Masked Diffusion Language Models*
  Role : PI
- Department of Energy, ($470,000 UVA portion : $100,000) 8/26-4/27
  Project title : *Genesis Mission Phase I : From Static to Adaptive Grids : Autonomous Topology Control at Scale*
  Role : co-PI
- Arena AI - contract, ($61,233) 7/26-2/27
  Project title : *Constrained-aware Diffusion for Optimized Chip Design*
  Role : PI
- 3Cavaliers Funding Award, ($80,000) 7/26-6/27
  Project title : *Personalized Post-Surgical Spine Care through Biomechanical Stress Profiling with AI-Powered Implantable Sensors*
  Role : co-PI (with Kyusang Lee (PI) and Jung-Bum Shin (co-PI))
- National Science Foundations, ($600,000) 9/25-8/28
  Project title : *Constrained Generation for Scientific and Engineering Applications*
  Role : PI

- **DARPA**, ($100,000) 8/25-7/26
  Project title : *Neuro-symbolic Generative Diffusion Models for Physically Grounded, Robust, and Safe Generation*
  Role : PI
- **Office of Naval Research**, ($180,000) 1/26-12/28
  Project title : *AI-SCORE : Artificial Intelligence School for Computer science and Operations Research Education*
  Role : PI (with Lavanya Marla (UIUC) as coPI)
- **4-VA**, 4-VA Collaborative Research Program ($30,000) 6/25-5/26
  Project title : *Towards Fair Decision Systems : Augmenting LLMs with Causal Graph Discovery*
  Role : PI (with Ziwei Zhu (GMU) as coPI)
- **4-VA**, 4-VA Collaborative Research Program ($5,000) 6/25-5/26
  Project title : *A Decomposition Architecture for Solving Constrained Combinatorial Optimization Problems*
  Role : co-PI (with Esra Toy (Virginia Tech) as lead PI)
- **NVIDIA**, NVIDIA Academic Research Award (GPUs usage) (15,000 hours on A100) 5/25-12/25
  Role : PI
- **Google**, Google Cloud Academic Research Award ($10,000 in compute credits) 4/25
  Role : PI
- **LaCross Institute**, 2025 Fellowship in AI Research ($100,000 [entire amount for the PI]) 06/25-05/27
  Role : PI (with collaborator Max Biggs)
- **Cohere for AI**, Cohere For AI Research Grant (LLM credits) ($20,000) 12/24
  Role : PI
- **University of Virginia (Research Innovation Award)** ($60,000) 8/24-7/25
  Project title : *Understanding and Mitigating Privacy Leakage Risks for Large Language Model Applications*
  Role : PI (with David Evans as coPI)
- **National Science Foundation (CISE - RI)** ($600,000 - UVA portion : $350,000) 08/23-07/26
  Project title : *Collaborative Research: End-to-end Learning of Fair and Explainable Schedules for Court Systems*
  Role : Lead PI (with L. Gouldin (Syr) as coPI and W. Yeoh WashU as collaborative PI)
- **National Science Foundation (EECS - EPCN)** ($520,000 - UVA porition : $260,000) 08/23-07/26
  Project title : *Collaborative Research: Physics Informed Real-time Optimal Power Flow*
  Role : PI (with Kyri Baker (UC Boulder) as collaborative PI)
- **Amazon Research Awards AWS AI** ($55,000) 01/23
  Project title : *Toward Understanding the Unintended Disparate Impacts of Private Machine Learning Systems*
  Role : PI
- **National Science Foundation (CAREER, CISE - RI)** ($515,403) 03/22-02/27
  Project title : *CAREER: End-to-end Constrained Optimization Learning*
  Role : PI
- **Google Research Scholar Award** ($60,000) 06/22
  Project title : *On the Equity of Differentially Private Decision Processes*
  Role : PI
- **National Science Foundation (CISE - SaTC)** ($500,000 - UVA portion : $281,000) 10/21-09/25
  Project title : *Collaborative Research: SaTC: Core: Small: Privacy and Fairness in Critical Decision Making*
  Role : Lead PI (with P. Van Hentenryck (Georgia Tech) as collaborative PI)
- **National Science Foundation (CISE - RI)** ($500,000 - UVA portion : $266,000) 10/20-09/24
  Project title : *Collaborative Research: RI: Small: Deep Constrained Learning for Power Systems*
  Role : PI (with P. Van Hentenryck (Georgia Tech) as collaborative PI)
- **CUSE Program** ($21,000) 07/21-06/23
  Project title : *On the Potential Perils of Fairness Algorithms in Decision Making and Learning Tasks*
  Role : PI (with S. Soundarajan (Syr) as coPI)

Travel and Service Grants

- *Support for Scholarship awards to attend the 2025 AAAI Privacy Preserving AI workshop* 03/25
  Sponsorship from : *Deloitte* ($3,000) ; *Google* ($5,000) ; *Apple* ($3,000) ; *OpenDP* ($500)
  Role : PI
- **National Science Foundation** ($50,000) 05/24
  Project title : *Conference : Artificial Intelligence Summer School for Computer Science and Operations Research Education*
  Role : coPI (with Lavanya Marla (UIUC) as PI

- **Artificial Intelligence Journal** ($4,000) 03/24
  Project title : *Student Support AI-SCORE 2024*
  Role : PI (with Lavanya Marla)
- **Artificial Intelligence Journal** ($15,000) 01/23
  Project title : *Student Support for AAMAS 2023*
  Role : PI (with Ana L. C. Bazzan)
- **National Science Foundation** ($25,000) 05/23
  Project title : *Travel : Travel : Doctoral Mentoring Consortium at the 22nd International Conference on Autonomous Agents and Multiagent Systems*
  Role : PI
- *Support for Scholarship awards to attend the 2024 AAAI Privacy Preserving AI workshop* 03/24
  Sponsorship from : *Google* ($5,000) ; *NJIT* ($2,500) ; *OpenDP* ($500)
  Role : PI
- *Support for Scholarship awards to attend the 2023 AAAI Privacy Preserving AI workshop* 03/23
  Sponsorship from : *Google* ($2,500)
  Role : PI

Tutorials, Selected Invited Talks and Media interviews

- **Invited Talk** (forthcoming ; title to be announced). Date TBA
  INFORMS Student Chapter, Virginia Tech
- **Keynote Talk** : Constrained-aware Generative AI for Scientific Applications. Oct 2026
  CS Research Symposium, Department of Computer Science, University of Virginia
- **Panel Chair and Moderator** : Who Does the Science ? Sep 2026
  ORNL AI-CORE Universities Workshop, University of Virginia
- **Panelist** : Genesis Mission Project Panel. Sep 2026
  ORNL AI-CORE Universities Workshop, University of Virginia
- **Invited Talk** : Grounding Generative Models in Constraints and Physical Principles : Towards Reliable Scientific and Engineering Designs. Aug 2026
  alphaXiv, AI for Science
- **Invited Talk** : Discrete LLMs for Scientific Research. July 2026
  DL4Sci Summer School, Berkeley, CA
- **Invited Talk** : Differentiable Optimization meets Generative Models. May 2026
  AI-SCORE Summer School
- **Invited Talk** : Research at the RAISE Lab. April 2026
  University of Virginia, UVA Department of Computer Science Town Hall
- **Invited Talk** : Constrained-aware Agents for Accelerating Scientific and Engineering Research. March 2026
  Biocomplexity, UVA
- **Invited Talk** : Constrained-aware Agents for Accelerating Scientific and Engineering Research. March 2026
  University of Michigan
- **Research Presentation** (with Max Biggs) : Privacy & Fairness in AI Pipelines. Jan 2026
  Fellowships in AI Research (FAIR) Symposium, LaCross Institute for Ethical AI in Business, UVA Darden School of Business
- **Invited Talk** : Speculative Diffusion Decoding : Massively Speeding up LLM inferences. Jan 2026
  Cerebras
- **Invited Talk** : Speculative Diffusion Decoding : Massively Speeding up LLM inferences. Jan 2026
  Capital One
- **Keynote Talk** : Foundations for Constrained-aware Diffusion Models. Dec 2025
  Constrained Optimization for Machine Learning (COML) Workshop at NeurIPS-25.
- **Invited Talk** : Generative AI with Constraints for Scientific Explorations. Nov 2025
  ORNL AI-CORE Workshop, Oak Ridge National Laboratory
- **Invited Talk** : Thrusting Generative AI for Scientific Discovery. Oct 2025
  CORE Workshop, Argonne National Laboratories
- **Keynote talk** : Physics-aware Models for Power Grids. Sep 2025
  Hexagon Workshop on Power Grids
- **Invited Talk** : Physics-aware Foundation Models for Scientific Discovery. Sep 2025
  AI initiative, Argonne National Laboratories

- **Invited talk** : Robust and Safety-constrained Foundation Models. Jul 2025
  The Trillion Parameter Consortium
- **Invited talk** : Constrained-diffusion Models for Physics-guided Discovery. Jul 2025
  USNCCM18
- **Invited talk** : Simultaneous Multi-Robot Motion Planning with Projected Diffusion Models. Jun 2025
  PloutosAI
- **Spotlight presentation** : Towards Physics-Aware Generative Inverse Solvers. Jun 2025
  DOE workshop on inverse design
- **Invited talk** : Generative AI for Science. Jun 2025
  TED-X Chantilly High School
- **Google talk** : Conversational audits for Privacy Exploits in LLM Agents. May 2025
  Google Research
- **Panelist** : Combining AI and ORMS for better trustworthy Decision Making. Mar 2025
  AAAI 2025 Bridge Program
- **Invited participant and group lead** : On the Safety of Foundations Models for Autonomous BioLabs. Feb 2025
  DOE Workshop on Envisioning Frontiers in AI and Computing for Biological Research
- **Keynote talk** : Privacy and Fairness issues in Large Language Models. Nov 2024
  S-HPC Workshop, at Supercomputing 24
- **Invited talk** : Unfairness in Constrained Machine Learning. Nov 2024
  Ohio State University, Department of Computer Science
- **Invited talk** : Constraining diffusion models for scientific applications. Oct 2024
  UVA LLM Workshop
- **Invited talk** : Privacy and Fairness in Resource Allocations. Oct 2024
  2024 Federal Committee on Statistical Methodology (FCSM) Research and Policy Conference
- **Invited talk** : Constrained Diffusion for Science and Engineering. Oct 2024
  Oklahoma State University, School of Industrial Engineering and Management
- **Invited talk** : Constrained Diffusion for Science and Engineering. Sep 2024
  University of Virginia, Department of Systems and Information Engineering
- **Podcast invited speaker** : NSI Cyber and Tech Center : "Unleashing Innovation : Navigating Game Changing Technologies" – episode on open source large language model. Jul 2024
  National Security Institute at George Mason University's Antonin Scalia Law School
- **Invited participant and group lead** : US-UK Scientific Forum on Science in the Age of AI. Jun 2024
  National Academy of Sciences
- **Panelist** : AI and OR summer school. May 2024
  AI-SCORE
- **Invited talk** : Fairness in ML : The curious case of computational shortcuts and hardware choices. May, 2024
  BuzzRobot
- **Invited talk** : The Principle of Data Minimization in Machine Learning. Apr, 2024
  Google Research Seminars
- **Media cover** : Building fairness into AI is crucial – and hard to get right. Mar 2024
  The Conversation , CHED/QR Radio
- **Invited talk** : Responsible AI in Decision Making Processes. Feb 2024
  Amazon Research Seminars
- **Keynote talk** : Privacy and Fairness in Societal Systems. Nov 2023
  Workshop on the Tradeoffs in Ethical AI, INRIA, France
- **Invited talk** : Responsible AI : Privacy and Fairness in Decision Making and Learning Tasks. Nov 2023
  TOC FOR FAIRNESS, Simons Collaboration on the Theory of Algorithmic Fairness
- **Panelist** : Navigating the Frontiers of Artificial Intelligence. Oct 2023
  The Center for Politics, University of Virginia
- **Invited talk** : Optimization and Learning for Science and Engineering. Oct 2023
  Conference on Complex Systems 2023
- **Invited talk** : ML for Optimization and Optimization for ML. Sep 2023
  AI/ML Seminar Series, University of Virginia
- **Keynote talk** : The Unintended Societal Effects of Privacy in Decision and Learning Tasks. Aug 2023
  IJCAI-2023, International Workshop on Mining Actionable Insights from Social Networks

- **Invited talk** : End-to-end Constrained Optimization Learning. Jul 2023
  AC Summer School : Machine Learning for Constraint Programming
- **Invited talk** : Differential Privacy for Power Systems. Jun 2023
  DTU PES Summer School
- **Invited talk** : Optimization Proxies and Differentiable Optimization for Decision Making. Jun 2023
  MARS Seminar, Pacific Northwest National Laboratory (PNNL)
- **Invited talk** : Constrained-aware Machine Learning in Energy Systems. Jun 2023
  IEEE Power and Energy Society webinar series
- **Invited talk** : Responsible AI : Privacy and Fairness in Decision and Learning Tasks. Apr 2023
  UC San Diego
- **Invited Talk** : Responsible AI : Privacy and Fairness in Decision and Learning Systems. Apr 2023
  Department of Computer Science and Engineering, Texas A&M University
- **Panelist** : ChatGPT : Charms and Challenges. Apr 2023
  Syracuse University
- **Invited talk** : Responsible AI : Privacy and Fairness in Decision and Learning Tasks. Mar 2023
  University of Virginia
- **Invited talk** : Constrained-Aware Machine Learning. Feb 2023
  Washington University in St. Louis
- **Invited talk** : Differential Privacy for Power Systems. Jan 2023
  Los Alamos National Lab's 5th Grid Science Winter School and Conference
- **Panelist** : Algorithmic Fairness and its Intersections. Dec 2022
  Thirty-sixth Conference on Neural Information Processing Systems (NeurIPS)
- **Tutorial** : End-to-end constrained optimization learning. Dec 2022
  21st International Conference of the Italian Association for Artificial Intelligence (AIxIA 2022)
- **Media cover** : How network pruning can skew deep learning models. Nov 2022
  Science Daily TechXplore AAAS EurekAlert
- **Invited talk** : Disparate Impacts in Privacy-preserving Machine Learning. Nov 2022
  Washington University in St. Louis
- **Tutorial** : Decision Focused Learning. Oct 2022
  Dagstuhl seminar on Data-Driven Combinatorial Optimisation
- **Media interview** *: Privacy and Fairness in AI.* *Jul/Sep 2022*
  Syracuse Media Report *NMSU News* *Sun News*
- **Media interview** *: Google Scholar Research Award.* *Jun 2022*
  Syracuse Media Report
- **Tutorial** *: Impacts of Data Privacy and Equity on Public Policy.* *Jun 2022*
  ACM Conference on Fairness, Accountability, and Transparency (FAccT)
- **Panelist** *: Fostering the Use of AI for Power System Transformation.* *Jun 2022*
  Climate Change AI
- **Media interview** *: NSF CAREER Award.* *Jun 2022*
  Syracuse Media Report
- **Invited talk** *: End-to-end constrained deep learning optimization.* *Mar 2022*
  Hall of Science (Kantar.com)
- **Panelist** *: AAAI-22 DC - Career Panel.* *Feb 2022*
  36th AAAI Conference on Artificial Intelligence (AAAI)
- **Invited talk** *: Privacy-preserving ML and decisions-making : uses and unintended disparate effects.* *Feb 2022*
  PriSec-ML (virtual seminars)
- **Media interview** *: AI for Climate Change.* *Dec 2021*
  RaiNews
- **Popular Media Report** *: ISSNAF Young Investigator Award.* *Nov 2021*
  New York Voice *AISE* Il Mattino StartupItalia Zox PugliaNews
- **Invited talk** *: Deep Constraint Learning : Applications and Privacy Considerations.* *Nov 2021*
  Italian Scientists & Scholars in North America Foundation
- ***Plenary Keynote talk*** *: Constraint-based Differential Privacy.* *Oct 2021*
  The International Conference on Principle and Practice of Constraint Programming (CP 2021) ,
- **Popular Media interview** *: Deep Learning for Engineering Applications.* *Nov 2021*
  Blum News

- ***Invited talk*** *: Privacy-Preserving Machine Learning : Uses and Unintended Disparate Effect.* *Sep 2021*
  ASPI Seminar (Syracuse University)
- ***Invited talk*** *: Differential Privacy and Machine Learning.* *May 2021*
  SUPA ECS workshop for High School Teachers
- **Invited talk** *: Deep Constraint Learning for Critical Engineering Systems.* *Nov 2020*
  Italian Scientists & Scholars in North America Foundation
- **Tutorial** *: Tutorial on Multiagent Optimization.* *Feb 2020*
  AAAI Conference on Artificial Intelligence (AAAI 2020)
- **Media cover** *: Multiagent Systems.* *Feb 2020*
  NetworkDigital360
- **Invited talk** *: Privacy-Preserving Artificial Intelligence.* *Jun 2019*
  University of Parma (CS Dept)
- **Tutorial** *: Tutorial on Multiagent Optimization for IoT Applications.* *May 2019*
  International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2019)
- **Invited talk** *: Differential Privacy for AI Applications*
  University of Southern California - Information Sciences Institute. *Jan 2019*
  Michigan State University. *Feb 2019*
- **Invited talk** *: Privacy Preserving Artificial Intelligence*
  Syracuse University. *Feb 2019*
  Drexel University. *Feb 2019*
  University of Arkansas. *Feb 2019*
  Colorado State University. *Mar 2019*
  University of Connecticut. *Mar 2019*
- **Tutorial** *: Tutorial on Constrained Multi-agent Optimization.* *Feb 2018*
  AAAI Conference on Artificial Intelligence (AAAI 2018)
- **Plenary Keynote talk** *: Distributed Constraint Optimization for Smart Energy Networks.* *Nov 2017*
  Italian Conference on Artificial Intelligence (AI*IA 2017)
- **Invited talk** *: Distributed Constraint Optimization*
  Delft University (TU Delft). *Apr 2016*
  University of Udine. *Apr 2016*
  New Mexico State University. *Mar 2016*
- **Invited talk** *: Large Neighboring Search for Distributed Constrained Optimization.* *Mar 2016*
  Ben-Gurion University of the Negev

Internal Service

School/Department Service (at UVA)

- Member, Engineering Research Advisory Council (ERAC)
- Chair, Faculty Search Committee (Tenure Track) 2026
- Member, AI Education Programs committee 2025
- Member, Faculty Search Committee (Teaching track) 2024 – 2025
- Member, Graduate Program Committee 2023 – 2025
- Advisor, ACM SIGAI at UVA 2023 – 2024

School/Department Service (at SU)

- Curriculum Committee 2023 – 2024
- Prepare and Grade Qualifier exam (Programming/Data Structure) 2022 – 2023
- Academic Integrity panelist 2021 – 2022
- Remembrance Scholars Selection Committee 2022

Professional Service

Conference Chair

- **International Conference on Principles and Practice of Constraint Programming (CP)** 2022
  with Roie Zivan

Workshop Chair

- **Sixth AAAI Workshop on Privacy Preserving Artificial Intelligence (PPAI)**, at AAAI 2025
  with Juba Ziani, Wanrong Zhang, and Jeremy Seeman
- **Algorithmic Fairness through the lens of Metrics and Evaluation (AFME)**, at NeurIPS 2024
  with Awa Dieng, Miriam Rateike, and Golnoosh Farnadi
- **AAAI Workshop on Learnable Optimization (LEARNOPT)**, at AAAI 2024
  with Elias B. Khalil, Pascal Van Hentenryck, Jan Drgona, Draguna Vrabie, and Priya Donti
- **Fifth AAAI Workshop on Privacy Preserving Artificial Intelligence (PPAI)**, at AAAI 2024
  with Juba Ziani, Christine Task, and Niloofar Mireshghallah
- **Algorithmic Fairness through the lens of Time (AFT)**, at NeurIPS 2023
  with Awa Dieng, Miriam Rateike, and Golnoosh Farnadi
- **Workshop on Optimization and Learning in Multi-Agent Systems**, at AAMAS 2023
  with Hau Chan, Jiaoyang Li, Filippo Bistaffa, and James Kotary
- **Fourth AAAI Workshop on Privacy Preserving Artificial Intelligence (PPAI)**, at AAAI 2023
  with Catuscia Palamidessi, and Pascal Van Hentenryck
- **Algorithmic Fairness through the lens of Causality and Privacy (AFCP)**, at NeurIPS 2022
  with Awa Dieng, Miriam Rateike, and Golnoosh Farnadi
- **Workshop on Optimization and Learning in Multi-Agent Systems**, at AAMAS 2022
  with Hau Chan and Jiaoyang Li
- **Third AAAI Workshop on Privacy Preserving Artificial Intelligence (PPAI)**, at AAAI 2022
  with Aleksandra Korolova and Pascal Van Hentenryck
- **AAAI Workshop on Machine Learning for Operational Research (ML4OR)**, at AAAI 2022
  with Emma Frejinger, Elias Khalil, and Pashootan Vaezipoor
- **Second AAAI Workshop on Privacy Preserving Artificial Intelligence (PPAI)**, at AAAI 2021
  with Pascal Van Hentenryck and Richard W. Evans
- **Workshop on Optimization and Learning in Multi-Agent Systems (OptLearnMAS)**, at AAMAS 2021
  with Amulya Yadev, Gauthier Picard, and Bryan Wilder
- **First AAAI Workshop on Privacy Preserving Artificial Intelligence (PPAI)**, at AAAI 2020
  with Pascal Van Hentenryck and Rachel Cummings
- **Workshop on Optimization and Learning in Multi-Agent Systems (OptLearnMAS)**, at AAMAS 2020
  with Bryan Wilder and Long Tran-Thanh
- **Workshop on Optimization in Multi-Agent Systems (OptMAS)**, at AAMAS 2019
  with Archie Chapman and Long Tran-Thanh
- **Workshop on Optimization in Multi-Agent Systems (OptMAS)**, at FAIM18 2018
  with Archie Chapman, Long Tran-Thanh, and Roie Zivan

Conference Organizing Committee

- **Steering Committee Chair** : The Second Summer School on AI and OR (AISCORE) 2026
- **Demo Chair** : International Conference on Autonomous Agents and Multiagent Systems (AAMAS) 2026
- **Steering Committee Chair** : The First Summer School on AI and OR (AISCORE) 2025
- **Tutorial Chair** : The ACM International Conference on AI in Finance (ICAIF) 2025
- **Demo Track Chair** : International Joint Conference on Artificial Intelligence (IJCAI) 2023
- **Scholarship Chair** : International Conference on Autonomous Agents and Multiagent Systems (AAMAS) 2023
- **Tutorial Chair** : International Conference on Autonomous Agents and Multiagent Systems (AAMAS) 2022
- **Track Chair** : International Conference on Principles and Practice of Constraint Programming (CP) 2018 – 2019
- **Track Chair** : International Conference on Autonomous Agents and Multiagent Systems (AAMAS) (For Learning and Adaptation track) 2025
- **Publicity Chair** : International Conference on Logic Programming (ICLP) 2019
- **Track Chair** : International Symposium on Mathematical Programming (ISMP) 2018

Award Committee

- ACP Early Career Researcher Award committee 2024 – 2026
- ISSNAF Mario Gerla Young Investigator Award 2023

Service to Journals

- **Associate Editor** : INFORMS Journal of Computing 2026–present
- **Associate Editor** : Journal of Artificial Intelligence Research (JAIR) 2025–present
- **Special Track Editor** : Journal of Artificial Intelligence Research (JAIR) *Special Track on Challenges in Pretraining* 2026–2027
- **Editorial Board Member** : Artificial Intelligence (AIJ) 2024–present
- **Associate Editor** : IISE Transactions *Special issue on Federated Learning* 2023
- **Guest Editor** : Theory and Practice of Logic Programming (TPLP) *Past and Present (and Future) of Parallel and Distributed Computation in (Constraint) Logic Programming* 2018

Senior Area Chair

- AAAI Conference on Artificial Intelligence (AAAI) 2026 – 2027
- ACM Conference on Fairness, Accountability, and Transparency (FAccT) 2023 – 2026
- International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS) 2026
- International Joint Conference on Artificial Intelligence (IJCAI) 2024 – 2025

Area Chair / Senior Program Committee

- Neural Information Processing Systems (NeurIPS) 2024 – 2026
- International Conference on Machine Learning (ICML) 2024 – 2026
- International Conference on Learning Representations (ICLR) 2026
- AAAI Conference on Artificial Intelligence (AAAI) 2020 – 2025
- International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS) 2023 – 2025
- International Joint Conference on Artificial Intelligence (IJCAI) 2021 – 2023
- European Conference on Artificial Intelligence (ECAI) 2023 – 2024
- International Conference on Principles and Practice of Constraint Programming (CP) 2018, 2019, 2022

Workshop/Tutorial Proposal Reviewer

- International Conference on Machine Learning (ICML) 2024 – 2025
- Neural Information Processing Systems (NeurIPS) 2023, 2024
- International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS) 2022

Program Committee

- Conference on Computer Vision and Pattern Recognition (CVPR) 2026
- ACM Computer and Communications Security (CCS) 2024 – 2026
- ACM Computer and Communications Security Asia (CCS Asia) 2026
- International Conference on Neuro-symbolic Systems (NeuS) 2026
- Bridge Program on AI and OR, at AAAI 2025
- IEEE Smartgridcomm 2025
- Neural Information Processing Systems (NeurIPS) 2020 – 2023
- International Conference on Machine Learning (ICML) 2021 – 2024
- International Conference on Learning Representations (ICLR) 2021 – 2025
- Privacy Enhancing Technologies Symposium (PETS) 2021 – 2023
- Electric Power System Research (PSCC) 2022
- International Conference on Logic Programming (ICLP) 2021
- International Conference on Principles and Practice of Constraint Programming (CP) 2016 – 2018, 2021
- International Joint Conference on Artificial Intelligence (IJCAI) 2016 – 2020
- European Conference on Machine Learning (ECML) 2020
- International Symposium on Combinatorial Search (SoCS) 2015 – 2020
- International Workshop on Optimization and Learning in Multi-Agent Systems (OptLearnMAS) 2020
- AAAI Conference on Artificial Intelligence (AAAI) 2018 – 2019
- Italian Conference on Computational Logic (CILC) 2017 – 2019
- Distributed Artificial Intelligence (DAI) 2019
- European Conference on Artificial Intelligence (ECAI) 2016 – 2018
- International Workshop on Optimization in Multi-Agent Systems (OptMAS) 2016 – 2017
- Italian Conference on Artificial Intelligence (AI*IA) 2017

Journal Reviewer

- Harvard Data Science Review — 2024
- INFORMS Journal on Computing — 2022, 2023
- Transactions on Machine Learning Research (TMLR) — 2022
- Journal of Artificial Intelligence Research (JAIR) — 2016 – 2022
- Artificial Intelligence Journal (AIJ) — 2016 – 2021
- Journal of Machine Learning Research (JMLR) — 2021
- IEEE Transactions on Smart Grid — 2019 – 2021
- IEEE Transactions on Power Systems — 2020 – 2021
- IEEE Transactions on Dependable and Secure Computing — 2020
- IEEE Transactions on Information Forensics & Security — 2019 – 2020
- Gates Open Research — 2020
- Patterns — 2020
- Autonomous Agents and Multi-Agent Systems (JAAMAS) — 2014 – 2017, 2019 – 2020, 2023
- Artificial Intelligence Review (AIR) — 2016 – 2017
- Fundamenta Informaticae Journal — 2016 – 2017
- AI Communications — 2017
- Algorithms for Molecular Biology (AMB) — 2014

Doctoral Consortia Mentoring

- AAAI Conference on Artificial Intelligence (AAAI) — 2022

Conference/Symposium/Workshop Reviewer

- European Control Conference (ECC) — 2021
- AAAI Conference on Artificial Intelligence (AAAI) — 2014 – 2017
- International Conference on Autonomous Agents and Multiagent Systems (AAMAS) — 2014 – 2016
- International Conference on Principles and Practice of Constraint Programming (CP) — 2016 – 2017
- International Conference on Principles and Practice of Multi-Agent Systems (PRIMA) — 2016
- International Joint Conference on Artificial Intelligence (IJCAI) — 2015
- International Conference on Logic Programming (ICLP) — 2015
- International Symposium on Combinatorial Search (SoCS) — 2014
- International Workshop on Distributed Constraint Reasoning (DCR) — 2014
- EURO-Par Parallel Processing (EUROPAR) — 2014
- Principles and Practice of Declarative Programming (PPDP) — 2014

Panel Reviewer

- NSF, CISE Panel ($\times 2$) — 2026
- NSF, CISE Panel — 2025
- ISSNAF TEF Research Grants — 2025
- AI Singapore Research - Governance Panel — 2025
- NSF, TIP Panel — 2025
- The Royal Society, Dorothy Hodgkin Fellowships — 2025
- NSF, CISE Panel ($\times 2$) — 2024
- Austrian Research Promotion Agency (FFG) — 2023
- NSF, Eng Panel — 2023
- NSF, NRT Panel — 2022
- NSF, SaTC Panel — 2022
- NSF, CISE Panel — 2022
- Israel Science Foundation (IIS) (external reviewer) — 2022 – 2023
- Climate Change AI (CCAI) Grant — 2022 – 2023
- CUSE Grant, Syracuse University — 2020 – 2021
- NSF, CISE RI (external reviewer) — 2020